

GPR Data Acquisition and Processing Guide

Lehrstuhl Rohstoff- und Ressourcenwirtschaft

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The idea of this guide is to serve as a quick reference throughout the setup of the equipment, as well as the acquisition and processing of GPR data. We will be using the IDS RIS MF Hi-Mod 1A georadar system (two antennas 200 & 600 MHz) , OneVision-1.0.1.90 beta software for data acquisition, and ReflexW 9.1.3 Sandmeier Software for data processing.

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- Data Processing

Materials and Information

Equipment

- Radar System RIS MF Hi Mod 1A
- Laptop Panasonic CF -31

Software

- OneVision-1.0.1.90 beta
- HD_DataConverter
- ReflexW 9.1.3 Sandmeier Software

Manual

- [RIS Hi Mod System User Manual](#)
- [One Vision 2.0 User Manual](#)
- [ReflexW Manual](#)
- [ReflexW Manual Booklet](#)

Online Contents

- [Sandmeier website](#) (how to use reflexw)
- [IDS Georadar - 4 Lite Setip and Scan](#) (a glimpse of OneVision configuration and field measurement)

Book Chapter

- [Cassidy - Ground Penetrating Radar Data Processing And Analysis](#) (guide about how to process gpr data)

1 - Equipment setup and cabling

The Radar System RIS MF Hi Mod 1A consists mainly of four components: antenna 200 & 600 MHz, DAD control unit, two batteries, and a Laptop Panasonic CF -31.

For a better understanding, check the [RIS Hi Mod System User Manual](#).

The final configuration used in the field can be seen in [our](#) video.

An internet video about how to set up the equipment can be watch here [IDS Georadar - 4 Lite Setup and Scan](#).

Next, a quick overview of the images showing how to connect the components:



Fig. 1. DAD sides. battery port and LAN port



Fig. 2. Lan cable connection between the Notebook Computer and the DAD Control Unit

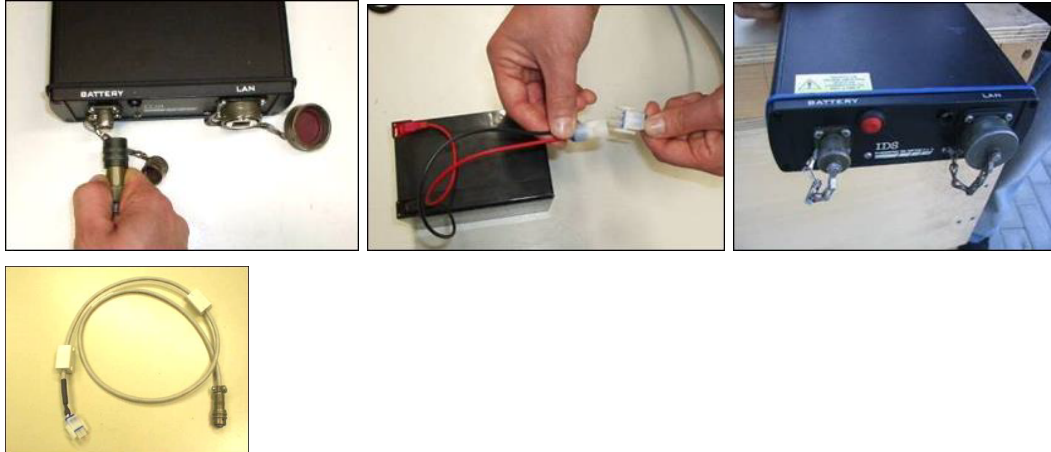


Fig. 3. Connecting the DAD Control Unit to the battery pack and switching on button



Fig4. DAD - Antenna connection & DAD - Wheel connection

2 - Software Configuration and Data Adquisition

We will be using OneVision-1.0.1.90 beta software to acquire the data. For a better understanding, check the [One Vision 2.0 User Manual](#). Also, an internet video about how to configure the software to scan can be found here [IDS Georadar - 4 Lite Setup and Scan](#).

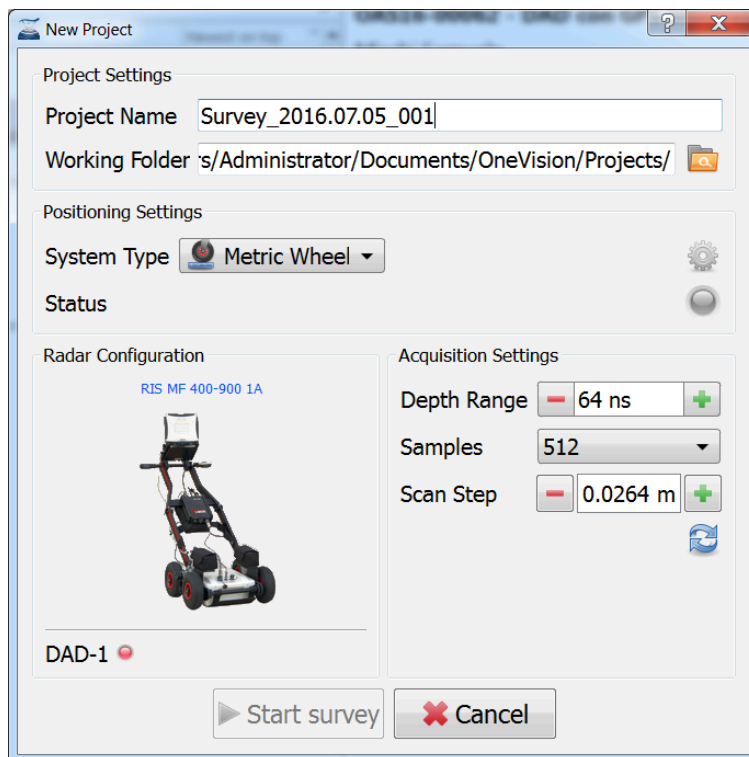
Open OneVision Software

- 1) Settings
 - a) Hardware → Radar configuration → RIS MF 200-600 1A, click on the icon, check status.

- b) Hardware → Encoder resolution → Proceed → Ready → Start to begin calibration - Push or Pull the radar, Put the input distance measured with measuring tape - This will update the encoder resolution number.
- c) Hardware → Serial port configuration → here goes the gps connection. We wont be using one.
- d) Graphics→ Channel Palette → Grayscale
- e) International → Metric system

2) New Project

- a) Project Name: choose test day number line number Survey_ 2024.09.01
- b) Working folder: User/documents/OneVision/projects/testRainer/Alps
- c) System type: Metric Wheel – single or multiple straight scans not georeferenced.
- d) Radar Configuration - Check everything is working and the antennas are there
- e) Adquisition Settings



It is important to register the following parameters during the survey, they will be used in the processing stage.

Spatial domain parameters

- i) **Profile Length** (record this number once the surveyed line is finished)
 - ii) **Scan Step** = number of samples per distance = movement speed / spatial frequency
- For:
- fine details* aprox 1 cm - 5 cm archaeological studies or detecting small objects
 - coarse details* – 10 cm to 20 cm geological surveys or locating large structures.

Scan step would be the spatial resolution (lateral)

600 MHz Antenna:

- High resolution (small targets): 0.01 to 0.03 meters.
- Medium resolution (medium-sized targets): 0.03 to 0.05 meters.
- General exploration: 0.05 to 0.1 meters.

200 MHz Antenna:

- Medium-high resolution (large targets): 0.1 to 0.2 meters.
- Medium resolution (medium-sized targets): 0.2 to 0.3 meters.
- General exploration (large-scale): 0.3 to 0.5 meters.

iii) **Number of Traces** = profile length/scan step

Time domain parameters

i) **Range (ns)**

ii) **Number of Samples**

- check aliasing problem →

$$\text{sampling frequency} = \frac{\text{number of samples}}{\text{range}} \gg 2 * f_{MAX} = 2 * 600\text{MHz} = 1200\text{MHz}$$

example: range=70ns, samples=512 provides sampling frequency around 7.30 GHz, which is ok

iii) **Propagation speed**

$$\text{propagation speed} = \frac{c}{\sqrt{\epsilon_r}}$$

$$\text{depth} = \frac{\text{velocity} * \text{time}}{2} = \frac{\text{propagation speed} * \text{range}}{2} = \frac{c}{\sqrt{\epsilon_r}} * \frac{\text{range}}{2}$$

$$c = 30 \text{ cm/ns}$$

- Air $\epsilon_r = 1$ $v \approx 30 \text{ cm/ns}$
- Dry Sand $\epsilon_r = 3 \text{ to } 6$ $v \approx 12.2 \text{ cm/ns to } 17.3 \text{ cm/ns}$
- Wet Sand $\epsilon_r = 20 \text{ to } 30$ $v \approx 5.5 \text{ cm/ns to } 6.7 \text{ cm/ns}$
- Dry Soil $\epsilon_r = 4 \text{ to } 6$ $v \approx 12.2 \text{ cm/ns to } 15 \text{ cm/ns}$
- Wet Soil $\epsilon_r = 10 \text{ to } 15$ $v \approx 7.7 \text{ cm/ns to } 9.5 \text{ cm/ns}$
- Fresh Water $\epsilon_r = 80$ $v \approx 3.4 \text{ cm/ns to }$
- Concrete $\epsilon_r = 6 \text{ to } 12$ $v \approx 8.7 \text{ cm/ns to } 12.2 \text{ cm/ns}$
- Asphalt $\epsilon_r = 4 \text{ to } 10$ $v \approx 9.5 \text{ cm/ns to } 15 \text{ cm/ns}$

depths:

lets say that propagation speed is 0.15 m/ns (dry soil) and that "range" selected is 64 ns.

The depth would be

$\text{Depth} = 0.15 \text{ m/ns} \times 64 \text{ ns} = 9.6 \text{ m} = 4.8 \text{ meters}$
 $D = \frac{0.15 \text{ m/ns} \times 64 \text{ ns}}{2} = \frac{9.6 \text{ m}}{2} = 4.8 \text{ m}$
 $D = 0.15 \text{ m/ns} \times 64 \text{ ns} = 9.6 \text{ m} = 4.8 \text{ meters}$

3) Start Survey

Once the setting has been concluded the Operator can start the survey. To start the acquisition, select Start Survey (). The "Start survey" button is enabled only if the radsar system is connected and the positioning system is correctly configured

4) Colection Screen

- Bottom left, information about the system: Radar signal, battery, project size limit
- Top right, start collecting data.
- We can select different antennas to display.
- Targets can be marked, and hyperbola fitting can be performed. Check dielectric constant
- Stop acquiring data. In the working directory/folder, we will find two folders: Data and Logs, as well as a filename.svy (which is the one to open if we want to open a file in OpenVision).

----- **After data acquisition** -----

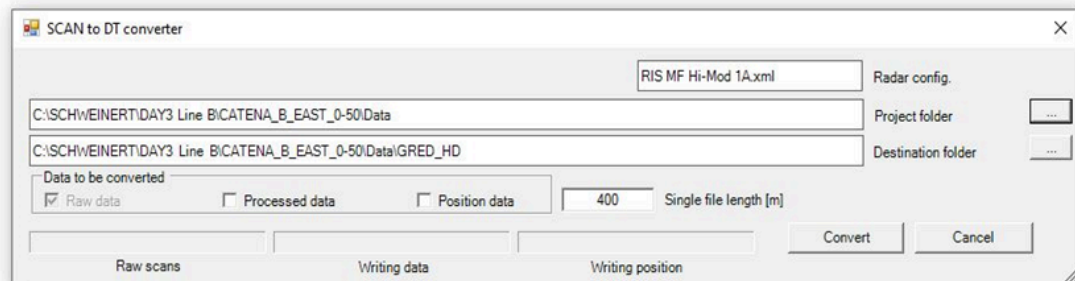
3- Data Conversion

Data Conversion From OneVision software to ReflexW software format

Convert the data using HD_DataConverter from one vision format to IDS format.

- Open the Hd_DataConverter → Select Project folder → Select the appropriate data folder (isin the Data folder) → then select the file (e.g., RIS MFF Hi-Mod 1A, XML document)

Name	Date modified	Type	Size
DataManagerNET.dll	29/03/2018 13:29	Application exten...	307 KB
Dforrt.dll	29/06/2016 14:54	Application exten...	444 KB
Geoconv2.dll	09/05/2018 18:30	Application exten...	17 KB
GPRConfiguration.dll	22/03/2018 14:19	Application exten...	789 KB
GPRConfigurationNET.dll	22/03/2018 14:19	Application exten...	20 KB
GPRFunctionF6.dll	29/06/2016 14:54	Application exten...	280 KB
GRED_Basic.dll	08/05/2018 18:24	Application exten...	145 KB
GredDXIF6.dll	29/06/2016 14:54	Application exten...	68 KB
GredProcesKernelF6.dll	31/03/2017 14:51	Application exten...	60 KB
HD_DataConverter	09/05/2018 18:30	Application	89 KB
HD_DataConverter	10/03/2022 17:42	Internet Shortcut	1 KB
Interpreter.dll	08/05/2018 18:24	Application exten...	32 KB
uninst	10/03/2022 17:42	Application	50 KB



- After running the program, a folder named GRED_HD is created inside the Data folder. Two files are generated with data from two different frequencies: LA010001.DT and LA020001.DT.

Check the log folder - contains information about calibration steps made by the equipment prior to the survey.

4 - Data Processing

Open ReflexW Software 9.1.3 (K J Sandmeier software)

- Create a folder where you are planning to work. → confirm project → That will create folders with different names inside the working directory.
- Open 2D data analysis

Step 1 Import data to Reflex

Open 2D data analysis → Import the data to ReflexW software

- File → Import → Input format (IDS) → DistanceDimen. (Meter) → Data type (Const. offset) → Convert to Reflex.
- → Data → GRED_HD folder → There are files of two different frequencies i.e. 600 MHz (LA010001.DT) and 200 MHz (LA020001.DT).
- The number indicated in yellow references to each frequency

Reflexw dataimport

Fileheader-coordinates DistanceDimen. METER data type const.offset ProfileDirection X ProfileConstant Y XStart: 0 XEnd: 51.39199972 YStart: 0 YEnd: 0 ZStart: 0 ZEnd: 0 number: 0		filename specification specification: original name prefix: filename factor: 1	
		Time and comment specification TimeDimension METER sample number: 512 time resampling: 1 time increment: 0.234375 timerange: 120 file header: 1024 trace header: 256	
format specification input format: IDS output format: new 16 bit integer scaling: 1		ConversionMode conversion sequence no max.traces/file: 1048576 line distance: 0 trace nr./2D-line: 0 trace incr.: 0 ☐ meandering ☐ check tracelength ☒ move receivers ☒ move shots	
ControlOptions ☒ control format ☒ read traceincr. ☐ fix endcoord. ☐ read coordinates ☐ read starttime ☐ read times from file ☐ read marker from file		☐ update traceheaders/gps coordinates ☐ use data folder update traceheaders: no utm-conversion: no ☐ calculate distancies ☐ correct for offset	
ControlPanel Convert to Reflex ☒ apply processing flow ☒ CheckExistingFiles ☐ check data for NAN ☒ PrimaryFile ☐ SecondaryFile			
		Exit Help Help	

- Now you have created in the folder ROHDATA files (.DAT). You can open these files whenever you want to work with the data

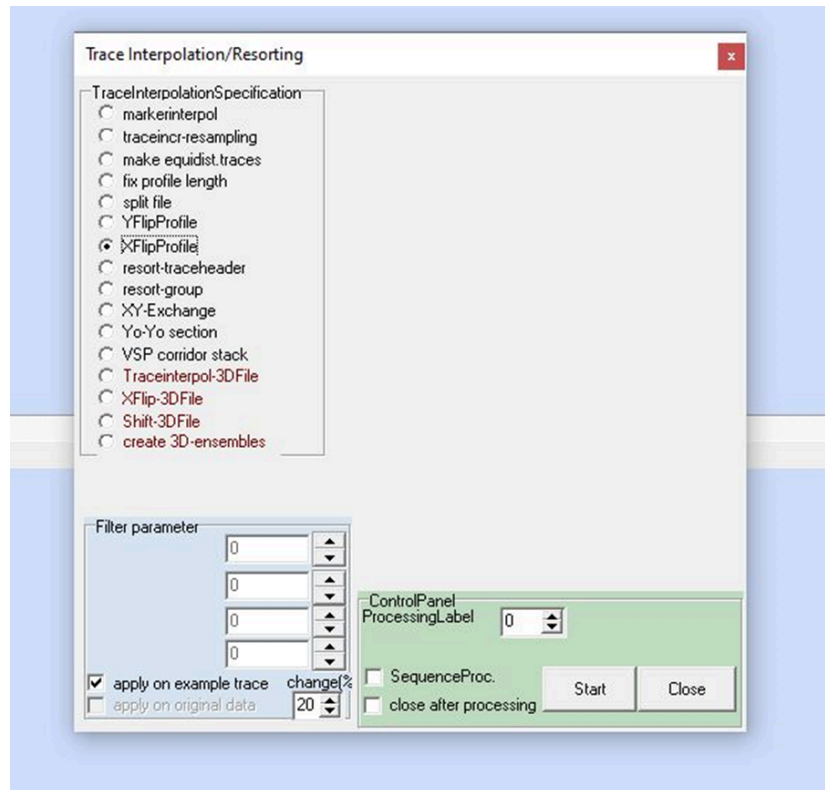
Once the file imported and converted to Reflex, a bunch of file are created:

- in the ROHData folder, a LA010001.DAT; LA1001.PAR; MESSROH.PAR
- in the main folder COMMENT.FIL; IMPORT.REP
- in a file with explanations of the configuration .
- Also you can export the jpg image in ascii

Step 2 X Flip

Only when necessary, flip the measurement line (in case you are moving forward and backwards along the same line)

- Processing → Trace interpolation/resorting → X FlipProfile
- Tick apply on example trace
- Change the processing label.



- Click 2->1 before proceeding to the next step

A file is going to be generated: Example: LA010001.01R & LA010001.01T

File→ Open→ PROCDATA→ LA010001.01T you will find the Flipped data.

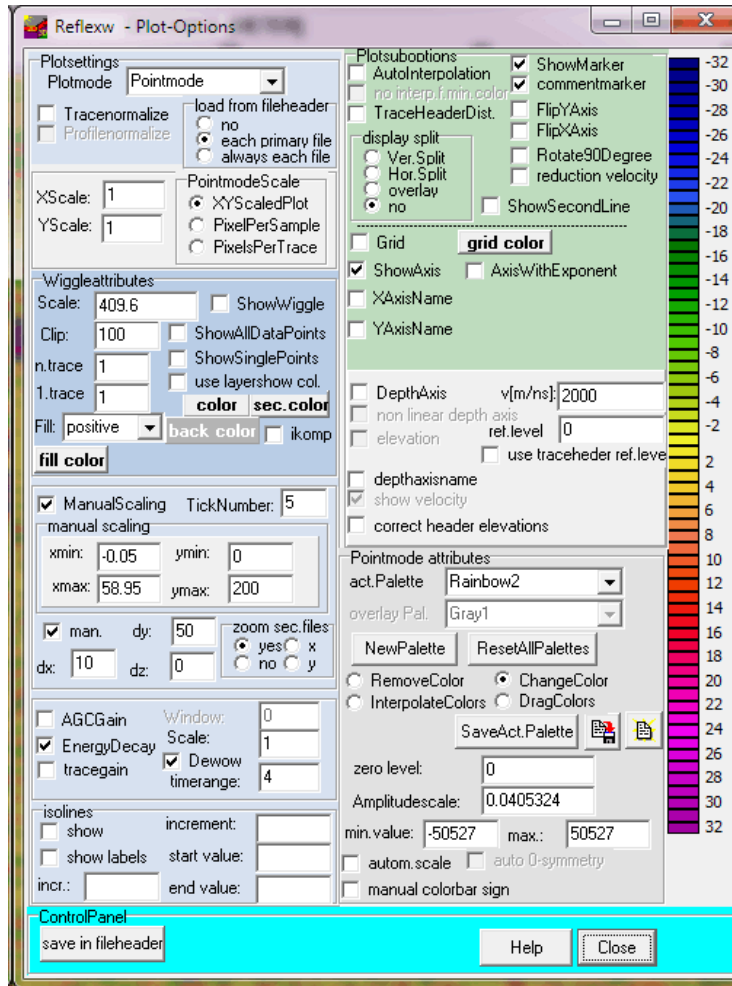
the number in yellow is going to be the processingLabel you chose.

Step: Export as image

File→ Export → Export format or copy image to clipBoard

Step 3 - Plot Options

After importing, the data is displayed using the standard plot options. You may change these plot options using the option Plot → Options.



The main plot options for GPR-data are:

- Plotmode
- PointmodeScale

xy the whole profile will be plotted to the actual window

pixel per sample – each data sample will be plotted by one screen pixel

pixel per trace means that one trace has the size of one pixel: useful for large data many traces

AmplitudeScale: this parameter controls the **contrast** of the data display.

Increase in Details: By decreasing the contrast scale, more subtle details become visible. This is because the range of intensity values is being expanded, allowing small variations to be better distinguished.

Darker Image: However, this can also make the overall image appear darker. Lower values become more prominent, causing areas with less reflection to appear black or darker.

Quick filtering:

- 1) **EnergyDecay**: most GPR-data are raw data without any gain in time-direction. Therefore activate this option if no gain-filter (see chap. II) has been applied on the GPR-data in order to compensate the energy decay with time.
- 2) **Dewow**: often GPR-data show a low-frequency content or DC-shift. With this option activated a running mean value is calculated for each value of each trace which is subtracted from the central point. Again use this option if no processing step like bandpassfiltering or dewow has been applied.

Normally the data must be filtered (see the filtering guide). The main filters are:

- static correction to compensate for the time delay of the first arrival.
- y(time)-gain
- dewow or bandpassfiltering
- clutter reduction (e.g. background removal)

Step 4: Filters → **DO NOT OVERPROCESS THE DATA**

summary:

- low frequency noise and high frequency noise, shall be eliminated using dewow and pass filtering
- shift time delay– the first arrival is not at time zero. perform time zero correction
- energy distribution is small, lower time range. therefore we must apply a time varying energy equalization, taking into account the theoretical diversion losses and normal losses due to attenuation
- clutter in horizontal band– eliminated using 2d filters
- migrations sometimes necessary in order to correct slopes of some dipping events.. concentrate the energy of diffractors
- topography correction

After applying a filter, two files are going to be generated in PRODATA folder: for example LA010001.05T & LA010001.05R

5 = processing step

T = trace

R= raw

so if you want to see/load the results after applying the filter, go for LA010001.05T

a) Dewow Filter

In Reflex: Processing → 1D filter → Select Subtract mean (dewow)

In GPR surveying, the transmitter and receiver are usually placed very close to each other. Near the transmitter, the fields have low-frequency components causing a slow temporal variation in the signal's mean level, known as "wow". This effect appears as a shift in the signal's mean level and can be removed by applying a high-pass filter, a process called "dewowing". This correction slightly darkens the image.

The goal of the dewow filter is to remove low-frequency noise while preserving the useful high-frequency data.

It is a high pass filter with a cutoff frequency

$$f_c = \frac{1}{T}$$

Example:

$T = 1.66\text{ns} \rightarrow f \approx 600\text{MHz}$ so dewow filter will erase frequencies below 600 MHz

$T = 10\text{ns} \rightarrow f \approx 100\text{MHz}$

$T = 20\text{ns} \rightarrow f \approx 50\text{MHz}$

$T = 33.3\text{ns} \rightarrow f \approx 30\text{MHz}$

$T = 100\text{ns} \rightarrow f \approx 10\text{MHz}$

un **time window de 3-5 ns**, que generará una frecuencia de corte de aproximadamente **200-300 MHz**, para la antena de 600
para la antena de 200, usamos 10 o 20 ns.

Frequencies below 100 MHz typically include unwanted noise such as system drift and environmental noise

- a) or also bandpass frequency → bandpass filter → bandpass frequency → order 3; butterworth;
lower cutoff, upper cutoff

REMEMBER TO CHANGE control Panel Processing Label each time a filter is applied.
top left window. 2 → 1 Band pass filter band to the left which is now the primary file.

b) Static correction

Get rid of the time delay. This is done using the option static correction - move to negative times.

Time zero correction to the first arrival, erase small artifacts

Processing → Static Correction →

static correction— move to negative times— choose in the trace

- a) static correction → interpolation— move to negative times. choose points by hand
- b) dynamic correction → receiver transmitter distance
- c) correct max phase → polarity positive, zero cross click, correct time auto. star window buscar inicio pico, end window buscar fin pico

Processing → Static Correction →

- d) move starttime → manual input — check starting time in the profile

c) **Gain correction**

Now we want to apply gain in order to enhance the signals from deeper underground.
increase the visualization the brightness of the reflectors at depth. Be careful to process the data, not over process it.

Processing → Gain →

- a) manual gain (y)
- b) gain function — choose test. By example: linear gain1, exponente 1, time start 0.1. also star time

d) **background removal**

The filter background removal under 2d filter allows to eliminate horizontal stripes.

Processing → 2D filter → background removal — use taper with 5. use start time higher than 5ns approx if you want to maintain the first arrival

e) **Bandpass filter**

may be used in order to get rid of high or low frequency noise.

Processing → 1D filter →

- a) bandpassfrequency
- b) bandpassbutterworth

butterworth ; zero padding

the filter requires the input of 4 parameters

f) Topographic correction

Processing → Static Correction → static correction

g) Migration

Migration improves the interpretation of GPR data by correcting diffraction, accurately relocating reflections, and enhancing image resolution, leading to clearer and more precise subsurface images.

Processing → migration → Diffraction stack

stacking – 60

Concentrates energy of diffractors